The Statistical Price of Directed Neighborhoods

Pahan Dewasurendra
Johns Hopkins University

Abstract

Conditional-average learning asks for the average target label in every graph neighborhood from i.i.d. labeled vertices. Its recent combinatorial characterization leaves a logarithmic sample- complexity gap: the upper bound contains $\alpha_2 \log(1/\epsilon)/\epsilon$, while the lower bound is only $\alpha_2/(\epsilon \log \alpha_2)$. We determine exactly when the accuracy logarithm is real.

For undirected neighborhoods, we prove that the original empirical-neighborhood learner has expected squared error at most

$$\frac{\alpha_1}{m+1} + \frac{7\alpha_2}{2m}.$$

The proof integrates its pointwise variance through a measure-valued weighted Caro–Wei inequality. This removes the logarithm without changing the algorithm, including on uncountable domains. Conversely, for directed neighborhoods we construct q disjoint transitive tournaments and a single known alternating concept for which every learner suffers $\Omega((q/m)\log(1+m/q))$ error with constant probability. The unknown object is only the marginal distribution. Independent mass imbalances hidden in unsampled pairs accumulate over nested suffixes, one harmonic scale at a time. A matching undirected lower bound follows from disjoint edges.

Consequently, at constant confidence the worst-case sample complexity for parameters $(\alpha_1, \alpha_2) = (d, q)$ is $\Theta((d+q)/\epsilon)$ for undirected graphs and $\Theta((d+q\log(1/\epsilon))/\epsilon)$ for directed graphs. A distribution-dependent neighborhood-leverage quantity explains both rates. Thus edge direction causes a genuine logarithmic statistical penalty even when the target concept is known, $\alpha_1 = 0$, and $\alpha_2 = 1$.

1 Introduction

Many learning tasks ask not for the label of one point, but for an average over points judged relevant to it. Examples include checking whether a local explanation is reliable, auditing subgroup behavior, and estimating preferences over a recommendation neighborhood (Ribeiro et al., 2018; Bhattacharjee & von Luxburg, 2024; Hardt et al., 2016). Bressan et al. (2026) recently isolated this task as *conditional-average learning*. A known directed graph G specifies a closed out-neighborhood $N[x]$ for every instance x. From i.i.d. samples labeled by an unknown binary concept c, a learner predicts

$$s_c(x) = \mathbb{E}_{X' \sim D}[c(X') \mid X' \in N[x]] \tag{1}$$

under integrated square loss.

Two joint graph–class parameters characterize learnability. The parameter α_1 is the largest independent set shattered by the concept class. For a fixed concept c, let B_c contain the vertices whose neighborhoods have both labels, and let $\alpha_2(G, c)$ be the independence number of the graph induced by B_c. The class parameter α_2 is the supremum over c. Finiteness of $\alpha_1 + \alpha_2$ is necessary and sufficient (Bressan et al., 2026). Their quantitative bounds, suppressing confidence for the moment, are

$$\Omega\left(\frac{\alpha_1 + \alpha_2/\log \alpha_2}{\epsilon}\right) \quad \text{and} \quad O\left(\frac{\alpha_1 + \alpha_2 \log(1/\epsilon)}{\epsilon}\right). \tag{2}$$

Table 1: Worst-case sample complexity at a universal constant confidence. Constants are universal. The directed logarithm is read as $1+\log(1/\epsilon)$.

Neighborhoods	Previous bounds	This work
Undirected	between (2)	$\Theta((\alpha_1+\alpha_2)/\epsilon)$
Directed	between (2)	$\Theta((\alpha_1+\alpha_2\log(1/\epsilon))/\epsilon)$

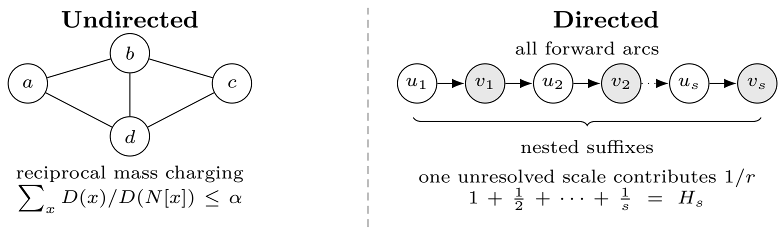

Figure 1: Why direction changes the rate. Symmetry makes inverse-neighborhood charges share one independence budget. A transitive directed chain supports nested suffixes at every mass scale, so unseen pair imbalances accumulate harmonically.

The upper logarithm arises by splitting neighborhood masses into dyadic levels. At every level, light neighborhoods have small total mass but their empirical means have large variance. It was unknown whether accumulating all levels is inherent.

We show that the answer is controlled by direction. In an undirected graph, symmetry couples the mass of every vertex to the masses of vertices that count it as a neighbor. A weighted independence inequality sums the inverse-neighborhood variances in one shot and eliminates all dyadic levels. Direction breaks this charging argument. A one-way chain has independence number one but has nested suffix neighborhoods at arbitrarily many mass scales. Independent uncertainties at those scales accumulate harmonically.

Contributions. Our results are summarized in Table 1.

First, we prove a sharp expected-risk upper bound for the existing learner on every measurable undirected graph:

$$\mathbb{E}L(h_S) \le \frac{\alpha_1}{m+1} + \frac{7\alpha_2}{2m}.$$

The central step is a weighted Caro–Wei inequality for probability measures. The finite weighted inequality is classical and has stronger power-weighted forms in feedback-graph learning (Caro, 1979; Wei, 1981; Eldowa et al., 2023). The novelty is that its direct integration of the conditional-average estimator removes the entire accuracy logarithm. An i.i.d. discretization argument extends the inequality to arbitrary measurable domains.

Second, we prove that the directed logarithm is unavoidable in its strongest qualitative regime. Our hard concept class is a singleton, so $\alpha_1 = 0$ and the learner knows every label before sampling. On q disjoint one-way chains, $\alpha_2 = q$. The unknown marginal puts a fixed total mass on each adjacent zero–one pair but hides an independent sign in its within-pair mass split. Not observing a pair reveals exactly no information about its sign. Every suffix average contains the sum of its missing signs. Averaging the resulting posterior variances over all suffixes produces the harmonic number H_s and risk $\Omega(q\log(1+m/q)/m)$. A Hilbert-space fourth-moment argument upgrades Bayes expectation to a constant-probability minimax lower bound.

Third, a separate Assouad construction on q undirected edges proves $\Omega(q/m)$ risk with a known concept. Combining these constructions with ordinary PAC hardness gives the exact worst-case envelopes in Table 1. We also introduce a distribution-dependent neighborhood leverage. It is at most α_2 for undirected graphs and equals $\Theta(qH_s)$ on the hard directed family, exposing the quantity that direction allows to grow.

2 Setting and prior result

Let $G = (\mathcal{X}, E)$ be a simple directed graph, with antiparallel edges allowed. Write $N^+(x) = \{y : (x,y) \in E\}$ and $N[x] = N^+(x) \cup \{x\}$. A set I is independent if no two distinct vertices in I have an edge in either direction. We assume the edge relation and all functions used below are measurable. Let $\mathcal{C} \subseteq \{0,1\}^{\mathcal{X}}$ and D be a distribution on $\mathcal{X}$. A sample $S = ((X_1, c(X_1)), \ldots, (X_m, c(X_m)))$ is drawn i.i.d. from D and an unknown $c \in \mathcal{C}$. The learner returns $h_S : \mathcal{X} \to [0,1]$ and incurs

$$L_D(h_S) = \mathbb{E}_{X \sim D}(h_S(X) - s_c(X))^2,$$

where s_c is defined in (1). Closed neighborhoods have positive mass almost surely under $X \sim D$, so the target is well defined where it is evaluated.

An independent set I is shattered if $|\{c|_I : c \in \mathcal{C}\}| = 2^{|I|}$. Let $\alpha_1(G, \mathcal{C})$ be the largest size of such a set. A neighborhood is *bichromatic* under c if it contains both labels. Let B_c be the set of its centers and $G_c = G[B_c]$. Define

$$\alpha_2(G, c) = \alpha(G_c), \qquad \alpha_2(G, \mathcal{C}) = \sup_{c \in \mathcal{C}} \alpha_2(G, c).$$

We abbreviate these parameters by α_1, α_2.

We use the learner of Bressan et al. (2026). If $S \cap N[x]$ is nonempty, it predicts the empirical fraction of ones there. Otherwise it invokes the one-inclusion-graph predictor on the vertices that are isolated in the graph induced by $S \cup \{x\}$. We need only two properties established in their proof. For every x,

$$\mathbb{E}_S(h_S(x) - s_c(x))^2 \leq \frac{7}{2mD(N[x])}, \tag{3}$$

and the integrated contribution of isolated monochromatic neighborhoods is at most $\alpha_1/(m+1)$. Nonisolated monochromatic neighborhoods incur zero loss. Their proof combines (3) with a light-neighborhood lemma separately on $O(\log(1/\epsilon))$ mass levels. We next avoid that partition when G is undirected.

3 Symmetry removes the accuracy logarithm

We first state the measure-valued inequality that allows all neighborhood masses to be charged at once. For an undirected graph $H = (V, F)$ and finite measure μ, write $\mu(N_H[x]) = \int \mathbf{1}\{y \in N_H[x]\}\, d\mu(y)$.

Lemma 1 (Measure-weighted Caro–Wei)**.** *Let H be a measurable undirected graph with finite independence number $\alpha(H)$ and let μ be a finite measure on its vertices. Then*

$$\int \frac{1}{\mu(N_H[x])}\, d\mu(x) \leq \alpha(H), \tag{4}$$

with the integrand set to zero on the null set where the denominator vanishes.

For a finite atomic measure, (4) is the weighted Caro–Wei inequality. Assigning independent exponential clocks with rates equal to the atom masses proves it: vertices whose clocks are smallest in their closed neighborhoods form an independent set, and the inclusion probability of x is its mass divided by its neighborhood mass. The non-atomic extension is not automatic, since one cannot assign an independent clock measurably to every point. Our full proof instead samples a finite graph from the measure and passes to a monotone limit. This also handles atoms and repeated sampled points, as shown in Appendix A.

Theorem 2 (Undirected expected risk)**.** *Let G be undirected and let h_S be the conditional-average learner above. For every D, $c \in \mathcal{C}$, and $m \geq 1$,*

$$\mathbb{E}_S L_D(h_S) \leq \frac{\alpha_1(G, \mathcal{C})}{m+1} + \frac{7\alpha_2(G, c)}{2m}. \tag{5}$$

Proof. Only bichromatic and isolated monochromatic neighborhoods can contribute. For $x \in B_c$, the closed neighborhood of x in G_c is contained in its closed neighborhood in G. Hence

$$D(N_G[x]) \geq D(N_{G_c}[x]).$$

Integrating (3) over B_c and applying Lemma 1 to the restriction of D to B_c gives

$$\int_{B_c} \mathbb{E}_S(h_S(x) - s_c(x))^2 dD(x) \leq \frac{7}{2m} \int_{B_c} \frac{dD(x)}{D(N_{G_c}[x])} \leq \frac{7\alpha(G_c)}{2m}.$$

The prior one-inclusion leave-one-out argument bounds the isolated monochromatic contribution by $\alpha_1/(m+1)$, and the remaining contribution is zero. ☐

The standard pointwise-median amplification in Bressan et al. (2026) turns any expected squared-loss bound r into error $O(r)$ with failure probability δ using $O(\log(1/\delta))$ independent runs. We therefore obtain the following immediate consequence.

Corollary 3 (Undirected sample complexity). *For undirected G, conditional averages are learnable with*

$$m(\epsilon, \delta) = O\left(\frac{\alpha_1 + \alpha_2}{\epsilon} \log \frac{1}{\delta}\right).$$

At any fixed constant confidence, the rate is $O((\alpha_1 + \alpha_2)/\epsilon)$.

The improvement is not a new algorithm. It is a structural fact about symmetric neighborhoods. In the original dyadic proof, every light-mass level can spend the full independence budget. In an undirected graph, Lemma 1 forces all levels to share one budget.

4 Direction forces harmonic accumulation

We now show that no analogous conclusion holds for directed graphs. The construction is distributional: the concept class contains one known function.

For integers $q, s \geq 1$, take q disjoint chains. Chain a is ordered

$$u_{a,1}, v_{a,1}, u_{a,2}, v_{a,2}, \ldots, u_{a,s}, v_{a,s}, \ldots,$$

and has an arc from every earlier vertex to every later vertex. There are no cross-chain arcs. Let $c(u_{a,k}) = 0$ and $c(v_{a,k}) = 1$, and set $\mathcal{C} = \{c\}$. Then $\alpha_1 = 0$. Every two vertices within a chain are adjacent in one direction. Every vertex except the last supported one has a later opposite-labeled neighbor. Choosing one such vertex per chain shows $\alpha_2 = q$.

Fix $\eta = 1/2$. For a sign array $\sigma \in \{-1, +1\}^{q \times s}$, define a marginal supported on the first s pairs by

$$D_\sigma(u_{a,k}) = \frac{1 + \eta\sigma_{a,k}}{2qs}, \qquad D_\sigma(v_{a,k}) = \frac{1 - \eta\sigma_{a,k}}{2qs}. \tag{6}$$

Every pair has sign-independent mass $1/(qs)$. At the first vertex of the suffix containing pairs $k, \ldots, s$, of length $r = s - k + 1$,

$$s_c(u_{a,k}) = \frac{1}{2} - \frac{\eta}{2r} \sum_{\ell=k}^{s} \sigma_{a,\ell}. \tag{7}$$

The cancellation in the denominator of (7) is the purpose of pairing.

Theorem 4 (Directed harmonic lower bound). *There are universal constants $c_0, c_1 > 0$ with the following property. For every $q \geq 1$ and $m \geq q$, the countable graph and singleton class above have $\alpha_1 = 0, \alpha_2 = q$, and for every possibly randomized learner there is a marginal D_σ such that*

$$\mathbb{P}_S\left\{L_{D_\sigma}(h_S) \geq c_0 \min\left(1, \frac{q}{m} \log\left(1 + \frac{m}{q}\right)\right)\right\} \geq c_1. \tag{8}$$

The concept and graph are known to the learner.

Proof idea. Set $s = \lceil m/q \rceil$ and give the signs in (6) independent Rademacher priors. We make the experiment easier by revealing $\sigma_{a,k}$ whenever either member of pair (a,k) is sampled. Pair occupancy is independent of its sign, so every missing pair retains an independent Rademacher sign.

Let $U_{a,r}$ be the number of missing pairs in the final r pairs of chain a. From (7), the posterior variance of the target at the first vertex of that suffix is $\eta^2 U_{a,r}/(4r^2)$. The mass of that vertex is at least $(1-\eta)/(2qs)$. Thus every predictor has conditional posterior risk at least

$$V = \frac{\eta^2(1-\eta)}{8qs} \sum_{a=1}^{q} \sum_{r=1}^{s} \frac{U_{a,r}}{r^2}. \tag{9}$$

If a pair lies t places from the chain end, its missing indicator has weight $w_t = \sum_{r=t}^{s} r^{-2}$. Since $\sum_t w_t = H_s$, and each pair is missing with probability $p = (1 - 1/(qs))^m \geq 1/16$, we have $\mathbb{E}V \geq cH_s/s$. Missing indicators are negatively correlated and $\sum_t w_t^2 = O(1)$, so Paley–Zygmund gives $V \geq cH_s/s$ with constant probability.

It remains to turn posterior variance into actual error for an arbitrary, possibly biased predictor. Restrict loss to the first vertex of every suffix and replace its sign-dependent mass by the lower bound $(1-\eta)/(2qs)$. Conditioned on observed data, this lower bound is $Q = \|A\xi - b\|^2$ in a fixed weighted Euclidean space, where ξ contains the missing independent signs. Hilbert-space Khintchine gives

$$\mathbb{E}Q^2 \leq 3(\mathbb{E}Q)^2. \tag{10}$$

Since $\mathbb{E}Q \geq V$, another Paley–Zygmund application yields $Q \geq V/2$ with conditional probability at least $1/12$. Averaging the sign prior then selects a deterministic σ. Appendix B gives all constants, handles nondivisibility, and proves (10).

For $q = 1$, the graph is a transitive tournament and both combinatorial dimensions are at most one. Nonetheless its sample complexity is $\Omega(\log(1/\epsilon)/\epsilon)$. This corrects the suppressed-log interpretation that tournaments are always $\widetilde{O}(1/\epsilon)$-easy: the tilde hides a necessary factor.

5 Matching envelopes and neighborhood leverage

The undirected upper rate is also necessary. The proof uses q disjoint edges and the same idea of hiding mass rather than labels.

Theorem 5 (Undirected lower bound)**.** *There are universal $c_2, c_3 > 0$ such that for every $q \geq 1$ and $m \geq q$, there is an undirected graph and singleton known concept with $\alpha_1 = 0, \alpha_2 = q$ for which every learner has some marginal D satisfying*

$$\mathbb{P}_S\{L_D(h_S) \geq c_2 q/m\} \geq c_3.$$

Take disjoint edges $\{u_i, v_i\}$, the known labels $c(u_i) = 0, c(v_i) = 1$, and $D_\sigma(u_i) = (1 + \eta\sigma_i)/(2q)$, $D_\sigma(v_i) = (1 - \eta\sigma_i)/(2q)$. The average on edge i is $(1 - \eta\sigma_i)/2$. Neighboring sign experiments have m-sample KL divergence at most $4m\eta^2/q$. Choosing η proportional to $\sqrt{q/m}$ and applying Assouad's lemma leaves a constant fraction of signs undecodable. Thresholding the learner's prediction at $1/2$ turns its square loss into a decoder loss, yielding $\Omega(q/m)$. Appendix C gives a constant-probability version.

For prescribed (d, q), append d isolated vertices on which the class realizes all binary labelings, while keeping it fixed on the hard component. This makes $\alpha_1 = d$ and $\alpha_2 = q$. A marginal supported on the isolated vertices gives the classical $\Omega(d/\epsilon)$ PAC lower bound (Ehrenfeucht et al., 1989). Supporting it on the hard component gives Theorem 4 or 5. The larger lower bound is at least half their sum.

Corollary 6 (Price of direction)**.** *Fix a universal confidence $1 - \delta_0$. Among problems with $(\alpha_1, \alpha_2) = (d, q)$, the worst-case sample complexity is*

$$\Theta\left(\frac{d+q}{\epsilon}\right) \quad \textit{for undirected graphs,} \qquad \Theta\left(\frac{d + q\log(1/\epsilon)}{\epsilon}\right) \quad \textit{for directed graphs.}$$

A distribution-dependent quantity explains the split. On a countable domain define

$$\Lambda_D(G, c) = \sum_{x \in B_c} \frac{D(x)}{D(N[x])}. \tag{11}$$

Equation (3) immediately gives

$$\mathbb{E} L_D(h_S) \leq \frac{\alpha_1}{m+1} + \frac{7\Lambda_D(G, c)}{2m}. \tag{12}$$

For undirected graphs, Lemma 1 says $\Lambda_D \leq \alpha_2$. On each hard directed chain, the mass of a suffix of r pairs is $r/(qs)$, while each first vertex has mass $\Theta(1/(qs))$. Hence $\Lambda_D = \Theta(qH_s)$. The lower bound matches (12) at the sample-dependent hard scale $s \asymp m/q$.

6 Related work

Conditional averages and local auditing. Bressan et al. (2026) introduced conditional-average learning, proved its exact learnability characterization, and obtained (2). Their lower bound already shows that marginal uncertainty can make a known concept hard, using bichromatic witnesses and degree binning. Our paired construction avoids degree binning and instead makes nested directed neighborhoods collect uncertainty at every scale. Their discussion notes that the light-neighborhood lemma is related to weighted Caro–Wei in the undirected case, but it does not derive the integrated inequality or the improved rate. Auditing local explanations also depends inversely on neighborhood locality (Bhattacharjee & von Luxburg, 2024). That task estimates one global auditing functional of supplied explanations, whereas conditional-average learning predicts every local mean.

Feedback graphs. Inverse-neighborhood sums are central in online learning with graph feedback. Directed graphs admit bounds with logarithmic factors (Alon et al., 2015). For undirected graphs, Eldowa et al. (2023) prove the stronger power-weighted inequality $\sum_v p(v)^{1+b}/p(N[v]) \leq \alpha^{1-b}$. Its $b = 0$ case contains finite weighted Caro–Wei and our combinatorial step. Their application is online regret and they identify directed feedback as substantially harder. Our contribution transfers the inequality to conditional-average risk on general measure spaces and proves that direction changes the minimax accuracy rate. Stochastic feedback graphs use different weighted independence and domination parameters based on edge-observation probabilities (Esposito et al., 2022).

Aggregate labels and conditional means. Learning from label proportions observes bag averages and predicts individual labels, reversing the information flow considered here. Recent work gives near-optimal variance-reduced ERM and SGD bounds in that setting (Busa-Fekete et al., 2025). Plug-in conditional-mean estimation and adaptive nearest-neighbor classification require geometric or function-class control absent from arbitrary neighborhood graphs (Grünewälder, 2018; Balsubramani et al., 2019).

7 Discussion

The separation isolates a structural cost, not an algorithmic artifact. Symmetry enforces reciprocal mass accounting. Direction permits many centers to charge the same downstream mass at different scales. The one-way chain is the extremal elementary example: its directed independence number is one, but its neighborhood leverage is harmonic.

Our exact envelopes hold at constant confidence. Combining Theorem 2 with median amplification gives the stated $\log(1/\delta)$ multiplier, while the existing general lower bound supplies $\Omega(\log(1/\delta)/\epsilon)$ (Bressan et al., 2026). We do not close the optimal joint dependence on vanishing δ. We also analyze squared loss. Other losses change the pointwise mean-estimation rate and may produce a different price for nesting.

The lower graph is deliberately order-like, not pathological in size or computation. Its neighborhoods encode queries such as "average outcome among people at least this old," an example already used to motivate directed conditional averages (Bressan et al., 2026). The result says that uniformly accurate local auditing of many nested threshold groups pays a logarithm even when the outcome function is public. If neighborhoods represent symmetric similarity instead, the logarithm vanishes.

Reproducibility statement

All claims are nonasymptotic and all proofs are included after the references. The supplementary artifact contains a short audit program. It exhaustively enumerates small paired-chain samples to check the posterior-variance identity, verifies the harmonic weights, enumerates the constant-probability occupancy step, checks the shifted Rademacher fourth-moment inequality, and tests weighted Caro–Wei against exact independence numbers on random small graphs. The program checks the derivations and is not used as an assumption.

References

Noga Alon, Nicolò Cesa-Bianchi, Ofer Dekel, and Tomer Koren. Online learning with feedback graphs: Beyond bandits. In *Proceedings of the 28th Conference on Learning Theory*, volume 40 of *Proceedings of Machine Learning Research*, pp. 23–35, 2015.

Akshay Balsubramani, Sanjoy Dasgupta, Yoav Freund, and Shay Moran. An adaptive nearest neighbor rule for classification. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Robi Bhattacharjee and Ulrike von Luxburg. Auditing local explanations is hard. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Marco Bressan, Nataly Brukhim, Nicolò Cesa-Bianchi, Emmanuel Esposito, Yishay Mansour, Shay Moran, and Maximilian Thiessen. Learning conditional averages. In *Proceedings of the 39th Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pp. 837–858, 2026.

Robert Istvan Busa-Fekete, Travis Dick, Claudio Gentile, Haim Kaplan, Tomer Koren, and Uri Stemmer. Nearly optimal sample complexity for learning with label proportions. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 5943–5976, 2025.

Yair Caro. New results on the independence number. Technical report, Tel Aviv University, 1979.

Andrzej Ehrenfeucht, David Haussler, Michael Kearns, and Leslie Valiant. A general lower bound on the number of examples needed for learning. *Information and Computation*, 82 (3):247–261, 1989.

Khaled Eldowa, Emmanuel Esposito, Tommaso Cesari, and Nicolò Cesa-Bianchi. On the minimax regret for online learning with feedback graphs. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Emmanuel Esposito, Federico Fusco, Dirk van der Hoeven, and Nicolò Cesa-Bianchi. Learning on the edge: Online learning with stochastic feedback graphs. In *Advances in Neural Information Processing Systems*, volume 35, pp. 25591–25602, 2022.

Steffen Grünewälder. Plug-in estimators for conditional expectations and probabilities. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pp. 1513–1521, 2018.

Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-precision model-agnostic explanations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

Victor K. Wei. A lower bound on the stability number of a simple graph. Technical report, Bell Laboratories, 1981.

A Measure-valued weighted Caro–Wei

We prove Lemma 1 without requiring a jointly measurable family of independent random clocks on an uncountable domain.

Proof of Lemma 1. The claim is invariant to scaling μ, so suppose first that μ is a probability measure. Draw $Z_1, \ldots, Z_k$ independently from μ. Construct a finite undirected graph on indices $[k]$ by joining $i \neq j$ when $Z_i = Z_j$ or Z_i and Z_j are adjacent in H. Any independent set of indices maps injectively to an independent set in H, so this sampled graph has independence number at most $\alpha(H)$.

The ordinary Caro–Wei inequality applied conditionally on the sampled graph gives

$$\sum_{i=1}^{k} \frac{1}{1+d_i} \leq \alpha(H),$$

where d_i is the degree of index i. Given $Z_i = x$, the closed-neighborhood count $1 + d_i$ has distribution $1 + Y$ for $Y \sim \mathrm{Bin}(k-1, p_x)$ and $p_x = \mu(N_H[x])$. Direct integration yields

$$\mathbb{E}\frac{1}{1+Y} = \int_0^1 (1 - p_x + p_x t)^{k-1} dt = \frac{1-(1-p_x)^k}{kp_x}.$$

Taking expectation, summing the k equal terms, and using measurability gives

$$\int f_k(p_x)\, d\mu(x) \leq \alpha(H), \qquad f_k(p) = \begin{cases} [1-(1-p)^k]/p, & p > 0, \\ k, & p = 0. \end{cases}$$

The functions f_k increase pointwise to $1/p$ for $p > 0$ and to infinity for $p = 0$. Monotone convergence therefore shows at once that the set $\{x : p_x = 0\}$ is μ-null and that the limiting integral is at most $\alpha(H)$. For a finite nonzero measure, normalize by its total mass. The zero measure is trivial. □

Remark 7. *For a finite atomic measure, an equivalent proof assigns an independent exponential clock of rate $\mu(x)$ to each positive-mass vertex. The set of vertices whose clock is smallest in its closed neighborhood is independent and has expected size equal to the left side of* (4)*.*

B Directed lower bound

We give a complete proof of Theorem 4. It suffices to consider deterministic learners because independent learner randomness can be conditioned upon throughout.

B.1 Posterior variance and harmonic weights

Fix q, m and set $s = \lceil m/q \rceil$, $n = qs$. Then $m \leq n < m + q \leq 2m$ when $m \geq q$. Use the marginal family (6) with $\eta = 1/2$ and put the uniform prior on its n signs. Augment the observation by revealing the sign of a pair whenever either endpoint is sampled. This can only decrease the minimax risk.

The total probability of every pair is $1/n$, independent of its sign. Thus the sequence of pair indices in the sample is i.i.d. uniform on $[n]$, independent of all signs. Conditional on these indices and the revealed signs, signs of unobserved pairs remain independent Rademachers.

For chain a, index suffixes by their number $r \in [s]$ of pairs. Let $U_{a,r}$ be the number of unobserved pairs in that suffix. Equation (7) shows that its random part at the first vertex is

$$-\frac{\eta}{2r} \sum_{j \in M_{a,r}} \sigma_j,$$

where $M_{a,r}$ is the missing set. Its posterior variance is $\eta^2 U_{a,r}/(4r^2)$. Also $D_\sigma(u_{a,k}) \geq (1-\eta)/(2n)$. Restricting the population loss to these first vertices proves the conditional variance lower bound (9).

Let

$$w_t = \sum_{r=t}^{s} \frac{1}{r^2}, \qquad Z = \sum_{a=1}^{q}\sum_{t=1}^{s} w_t I_{a,t},$$

where $I_{a,t}$ indicates that the pair t places from the end of chain a is missing. Exchanging the order of summation gives

$$\sum_{a=1}^{q}\sum_{r=1}^{s} \frac{U_{a,r}}{r^2} = Z, \qquad \sum_{t=1}^{s} w_t = \sum_{r=1}^{s} \frac{r}{r^2} = H_s. \tag{13}$$

Furthermore $w_t \leq t^{-2} + \int_t^\infty x^{-2}dx \leq 2/t$, hence $\sum_t w_t^2 \leq 2\pi^2/3$.

Every pair is absent with probability $p = (1-1/n)^m$. For $n \geq m \geq 1$, $p \geq 1/4$ except when $n = m = 1$. For distinct pairs,

$$\mathbb{E}[I_i I_j] = (1-2/n)^m \leq (1-1/n)^{2m} = p^2,$$

so their covariance is nonpositive. Consequently

$$\mathbb{E}Z = qpH_s, \qquad \mathbb{E}Z^2 \leq (\mathbb{E}Z)^2 + pq\sum_t w_t^2 \leq (\mathbb{E}Z)^2 + \frac{2\pi^2 pq}{3}.$$

Since $q \geq 1$, $p \geq 1/4$, and $H_s \geq 1$, Paley–Zygmund implies

$$\mathbb{P}\{Z \geq \mathbb{E}Z/2\} \geq \frac{1}{4}\frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2} \geq c_{\text{occ}} \tag{14}$$

for a universal $c_{\text{occ}} > 0$. On this event,

$$V \geq \frac{\eta^2(1-\eta)p}{16}\frac{H_s}{s} \geq c\frac{H_s}{s}. \tag{15}$$

B.2 A shifted Rademacher small-ball lemma

Lemma 8. *Let $\xi_1, \ldots, \xi_N$ be independent Rademachers, let $v_1, \ldots, v_N, b$ lie in a real Hilbert space, and put $Y = \sum_i \xi_i v_i$ and $Q = \|Y - b\|^2$. Then*

$$\mathbb{E}Q^2 \leq 3(\mathbb{E}Q)^2, \qquad \mathbb{P}\{Q \geq \mathbb{E}Q/2\} \geq 1/12.$$

Proof. Write $v = \mathbb{E}\|Y\|^2 = \sum_i \|v_i\|^2$ and $B = \|b\|^2$. The Hilbert-space fourth-moment inequality gives $\mathbb{E}\|Y\|^4 \leq 3v^2$. One direct proof expands the fourth power: terms with an index of odd multiplicity vanish, while each surviving cross term is bounded by the corresponding product of squared norms. Also $\mathbb{E}\langle Y, b\rangle^2 = \sum_i \langle v_i, b\rangle^2 \leq vB$. Symmetry makes all cubic terms vanish. Expanding Q^2 therefore gives

$$\mathbb{E}Q^2 = \mathbb{E}\|Y\|^4 + B^2 + 4\mathbb{E}\langle Y, b\rangle^2 + 2Bv \leq 3v^2 + B^2 + 6Bv \leq 3(v+B)^2.$$

Since $\mathbb{E}Q = v + B$, the first claim follows. Paley–Zygmund applied to the nonnegative variable Q with threshold $1/2$ gives the second. □

Condition on sample pair indices, revealed signs, and learner randomness. The vector of all target values at the first suffix vertices is an affine image of the missing independent signs. Define Q by weighting the squared errors on those vertices uniformly by the lower bound $(1-\eta)/(2n)$. The actual population error dominates Q, while $Q = \|A\xi - b\|^2$ in a fixed weighted Euclidean space. Its conditional expectation is at least the integrated posterior variance V. By Lemma 8, $Q \geq V/2$ with conditional probability at least $1/12$. Combining with (14) and (15) gives joint probability at least $c_{\text{occ}}/12$ that

$$L_{D_\sigma}(h_S) \geq c\frac{H_s}{s}.$$

This probability is averaged over the uniform sign prior. Hence some deterministic σ has the same frequentist lower bound. Finally, $H_s/s \geq c\min\{1,(q/m)\log(1+m/q)\}$ because $s = \lceil m/q \rceil$. This proves Theorem 4 except at $q = m = 1$. At that endpoint, use the same two marginals on the first pair without revealing its sign. Their one-sample total variation is $\eta < 1$, while their conditional averages differ by η. Le Cam's two-point argument gives a universal constant-probability constant-risk lower bound, which is the required right side of (8) after changing constants.

B.3 Inverting the risk lower bound

For completeness, the condition $m < cq\log(1/\epsilon)/\epsilon$ with a sufficiently small universal c implies $(q/m)\log(1+m/q) > c'\epsilon$ for all sufficiently small ϵ. Bounded values of ϵ are absorbed by changing constants. Thus Theorem 4 implies $\Omega(q\log(1/\epsilon)/\epsilon)$ constant-confidence sample complexity.

C Undirected lower bound

We prove Theorem 5. For the nontrivial regime $m \geq q$, fix $\eta = a\sqrt{q/m} \leq 1/2$ for a sufficiently small universal a. Put the uniform prior on $\sigma \in \{-1,+1\}^q$. Recall that the two endpoints of edge i both have target $\theta_i = (1-\eta\sigma_i)/2$.

For sign arrays differing only at coordinate i, the one-sample KL divergence is

$$\mathrm{KL}(D_\sigma \| D_{\sigma^{(i)}}) = \frac{\eta}{q}\log\frac{1+\eta}{1-\eta} \leq \frac{4\eta^2}{q}.$$

Tensorization and Pinsker give

$$\mathrm{TV}(D_\sigma^m, D_{\sigma^{(i)}}^m) \leq \sqrt{2m\eta^2/q} \leq \sqrt{2}a.$$

Choose $a \leq 1/(4\sqrt{2})$. Assouad's argument then shows that for every decoder $\widehat{\sigma}$,

$$\mathbb{E}_{\sigma,S} d_H(\widehat{\sigma},\sigma) \geq \frac{3q}{8}. \tag{16}$$

Given a predictor h, decode $\widehat{\sigma}_i = +1$ if the average of $h(u_i)$ and $h(v_i)$ is below $1/2$, and -1 otherwise. If this decoder errs, convexity gives

$$(h(u_i)-\theta_i)^2 + (h(v_i)-\theta_i)^2 \geq \frac{\eta^2}{2}.$$

Every endpoint has mass at least $(1-\eta)/(2q) \geq 1/(4q)$, so

$$L_{D_\sigma}(h) \geq \frac{\eta^2}{8q} d_H(\widehat{\sigma},\sigma). \tag{17}$$

Equations (16)–(17) give Bayes expected risk at least $3\eta^2/64 = \Omega(q/m)$.

To obtain a probability statement, one may use a direct bounded-Hamming argument. Since $0 \leq d_H \leq q$ and its mean is at least $3q/8$,

$$\mathbb{P}\{d_H \geq 3q/16\} \geq \frac{(3/8)-(3/16)}{1-(3/16)} = \frac{3}{13}.$$

Together with (17), the joint prior-and-sample probability of risk $\Omega(q/m)$ is at least $3/13$. Averaging selects one deterministic σ. If $m < q$, set η to a small constant and apply the same proof to obtain constant risk, completing the stated $\min\{1, q/m\}$ form.

D Auxiliary audits

The supplementary program implements the following independent checks.

1. It enumerates every sample-location sequence for small (q, s, m) and confirms $\mathbb{E}V = pH_s/(4s)$ for the simpler uniform-label version of the chain calculation.
2. It verifies $\sum_t w_t = H_s$ and the uniform square-sum bound through $s = 999$.
3. It enumerates all occupancy patterns induced by samples for small chains and checks the constant-probability threshold in (14).
4. It enumerates Rademacher signs for random linear maps and shifts and checks $\mathbb{E}Q^2 \leq 3(\mathbb{E}Q)^2$.
5. It compares the weighted Caro–Wei sum with an exact exhaustive computation of the independence number on random undirected graphs up to eight vertices and weights spanning six orders of magnitude.
6. It exhausts the actual paired marginals and verifies their normalization, suffix targets, posterior-variance identity, and the neighboring KL formula used in the edge lower bound.